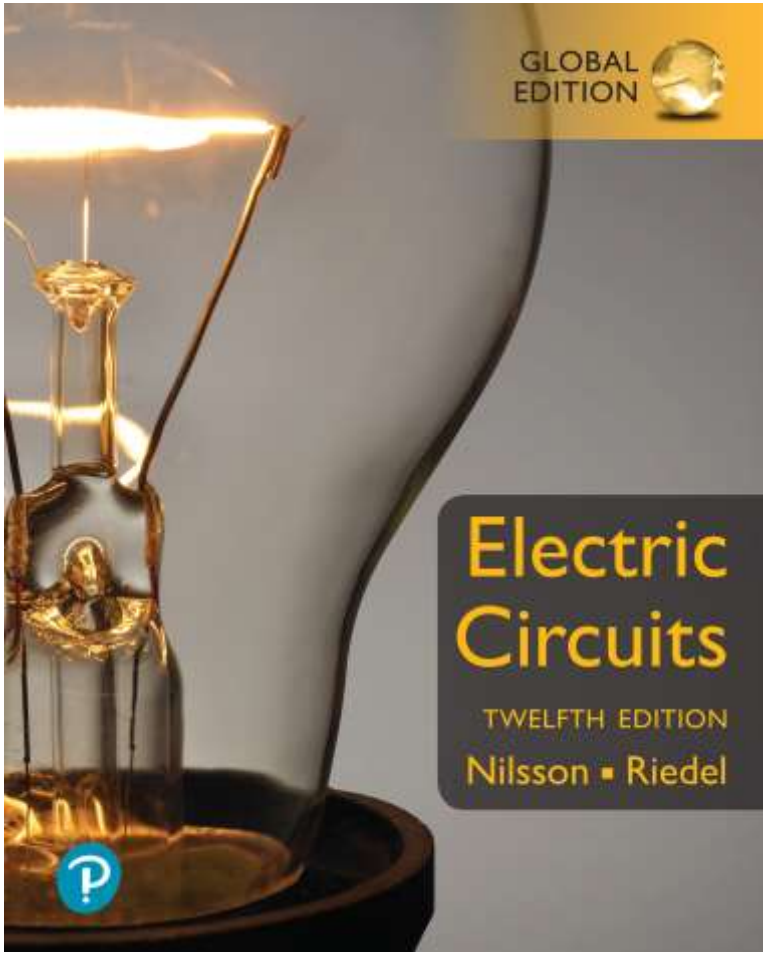
Electric Circuits

Twelfth Edition, Global Edition



Chapter 3 – Part 2

Simple Resistive Circuits

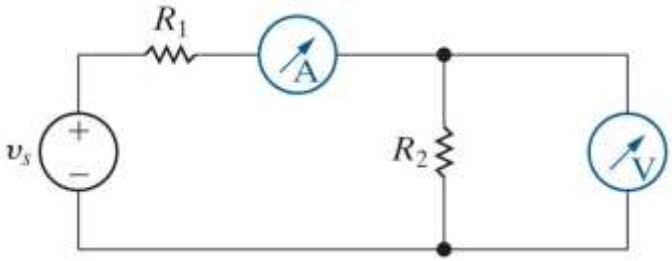
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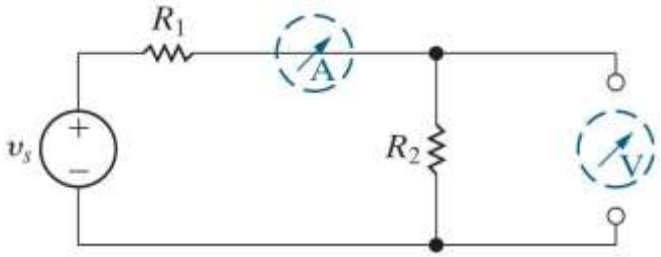
Chapter 3 – Simple resistive circuits (1 of 22)

Measuring current and voltage

- An ammeter is an instrument that measures current. It is placed in series with the circuit element whose current is being measured. Ideally, the ammeter introduces no resistance in the circuit, so behaves like a short circuit.
- A voltmeter is an instrument that measures voltage. It is placed in parallel with the circuit element whose voltage is being measured. Ideally, the voltmeter behaves like an open circuit.



An ammeter measuring R_1 current and a voltmeter measuring R_2 voltage.



Models for an ideal ammeter and ideal voltmeter.

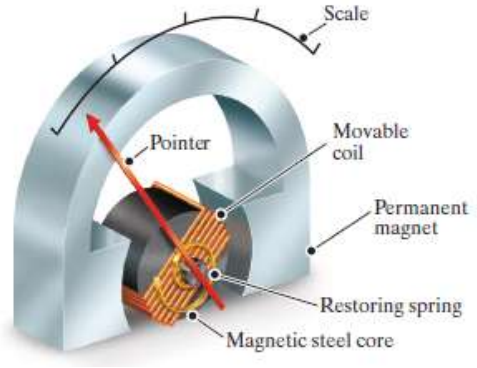
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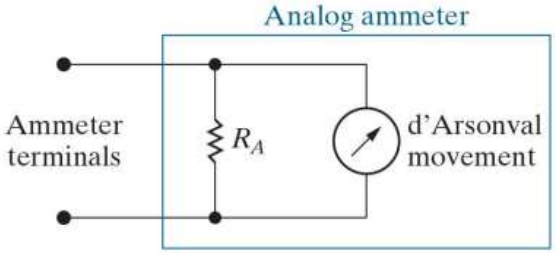
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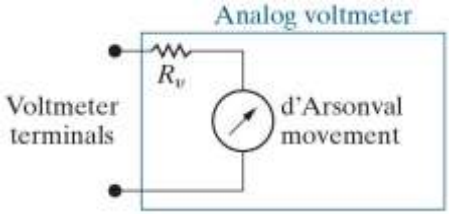
Chapter 3 – Simple resistive circuits (2 of 22)

Measuring current and voltage

- Meters used in modern laboratories are usually digital. It is instructive, however, to examine analog meters, as they provide a practical application of the circuit analysis techniques we have introduced thus far.
- Analog meters are based on the d'Arsonval meter movement. The moveable coil in the meter movement has both a voltage rating and a current rating. If a meter movement is rated at 50 mV and 1 mA, then when the coil carries 1 mA its voltage drop is 50 mV and the pointer deflects to its full-scale reading.







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Chapter 3 – Simple resistive circuits
(3 of 22)

Measuring current and voltage

- The parallel resistor in the analog ammeter limits the current in the moveable coil (because of current division), while the series resistor in the analog voltmeter limits the voltage across the moveable coil (because of voltage division).
- Analog meters are non-ideal. Both the added resistor and the meter movement introduce resistance in the circuit where the meter is attached. How much the meters alter the circuit being measured depends on the effective resistance of the meter compared with the resistance in the circuit.

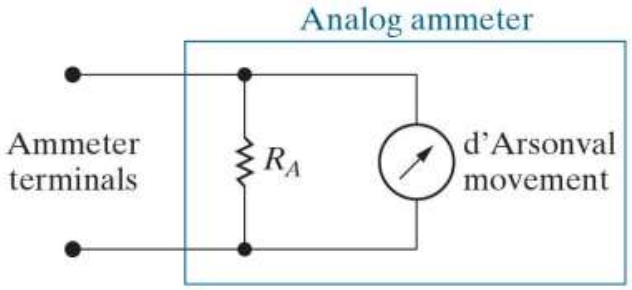
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Chapter 3 – Simple resistive circuits (4 of 22)

Measuring current and voltage – Example:
A 50 mV, 1mA D'Arsonval movement is used in a ammeter with a full-scale reading of 10 mA.

- Find R_A.
- Find the resistance that is added to a circuit when this ammeter is used to measure current.

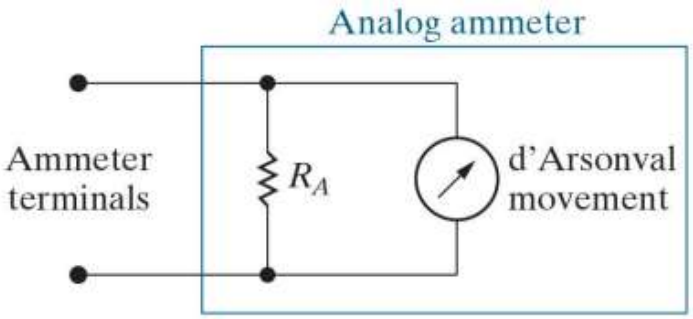


The current in the ammeter divides between the R_A resistor and the meter movement. When the current in the ammeter is 10 mA, the coil current is 1 mA and the resistor current is 9 mA. When the coil current is 1 mA, its voltage is 50 mV.

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Chapter 3 – Simple resistive circuits
(5 of 22)

When the coil current is 1 mA, its voltage is 50 mV. What is the voltage across the R_A resistor?



A. 1/50 V

B. 50 V

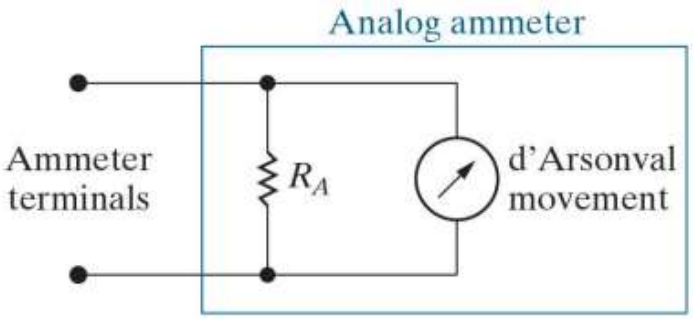
C. 50 mV

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Chapter 3 – Simple resistive circuits
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When the coil current is 1 mA, its voltage is 50 mV. What is the voltage across the R_A resistor?



- ✗ A. 1/50 V
- ✗ B. 50 V
- ✓ C. 50 mV

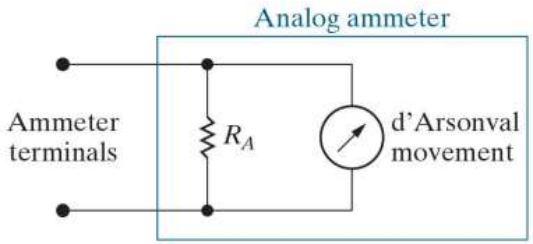
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Chapter 3 – Simple resistive circuits
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Measuring current and voltage – Example (continued):
A 50 mV, 1mA D'Arsonval movement is used in a ammeter with a full-scale reading of 10 mA.

- Find R_A.
- Find the resistance that is added to a circuit when this ammeter is used to measure current.



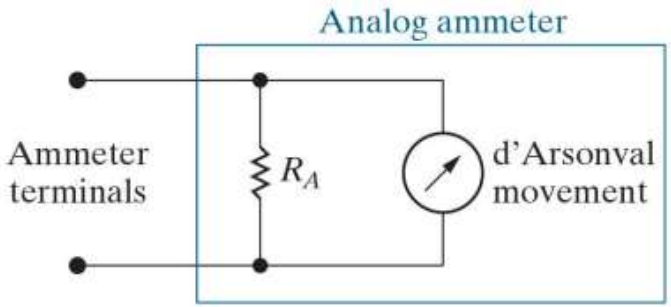
From Ohm's law, $50\times10^{-3} = R_A\left(9\times10^{-3}\right)$

$$\Rightarrow \qquad R_A = \frac{50\times10^{-3}}{9\times10^{-3}} = 5.555\ \Omega$$

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Chapter 3 – Simple resistive circuits
(7 of 22)

If a d'Arsonval meter movement is rated 50 mV and 1 mA, what is its resistance?



A. $50\,\Omega$

B. $1/50\,\Omega$

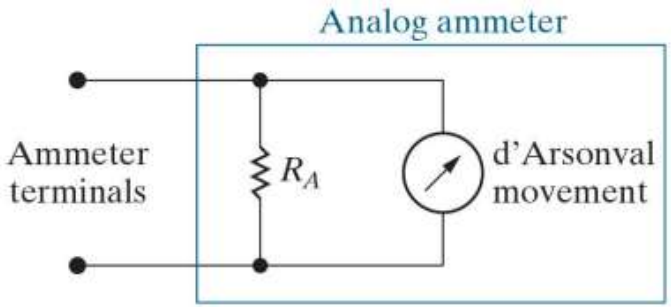
C. $50\ \text{m}\Omega$

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Chapter 3 – Simple resistive circuits
(7 of 22)

If a d'Arsonval meter movement is rated 50 mV and 1 mA, what is its resistance?



✔ A. 50 Ω

✘ B. 1/50 Ω

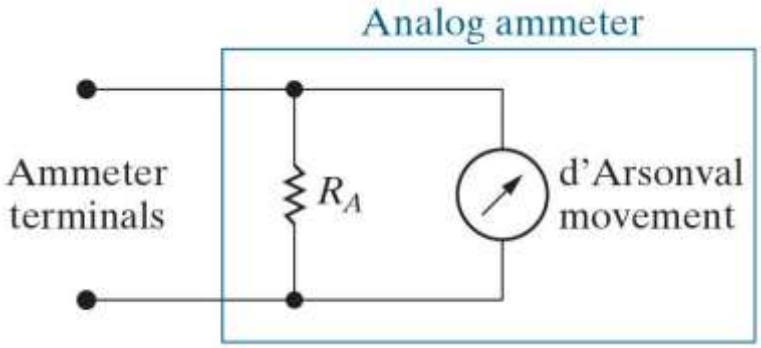
✘ C. 50 mΩ

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Chapter 3 – Simple resistive circuits
(8 of 22)

To find the effective resistance of the ammeter, combine the RA resistance and the d'Arsonval movement resistance in



A. series

B. parallel

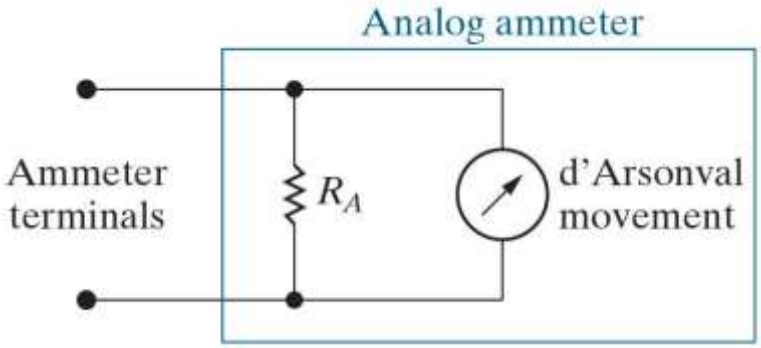
C. These resistances cannot be combined

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Chapter 3 – Simple resistive circuits
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To find the effective resistance of the ammeter, combine the RA resistance and the d'Arsonval movement resistance in



✗ A. series

✓ B. parallel

✗ C. These resistances cannot be combined

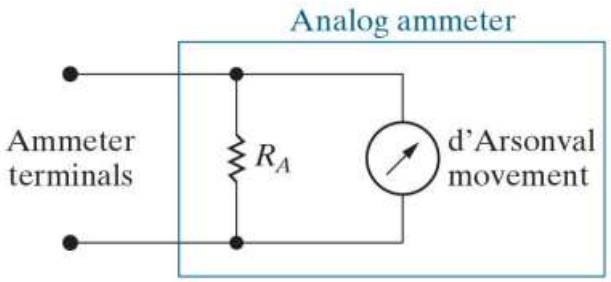
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Chapter 3 – Simple resistive circuits
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Measuring current and voltage – Example (continued):
A 50 mV, 1mA D'Arsonval movement is used in a ammeter with a full-scale reading of 10 mA.

- Find R_A.
- Find the resistance that is added to a circuit when this ammeter is used to measure current.



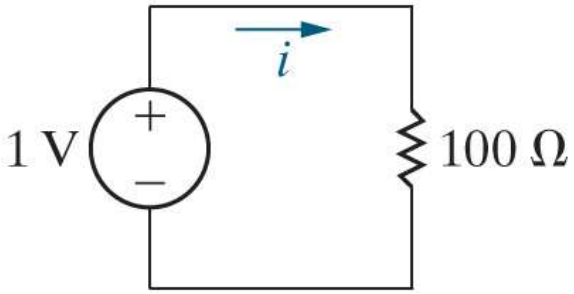
The resistance of the ammeter is $R_m = R_A \,||\, R_{\text{d'Arsonval}} = 5.555 \,||\, \left(50 \times 10^{-3}\right)$

$$= 5\ \Omega$$

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Chapter 3 – Simple resistive circuits
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If we want to measure the current *i* in the circuit shown here, we should insert an ammeter in _____ with the 100 Ω resistor.



A. series

B. parallel

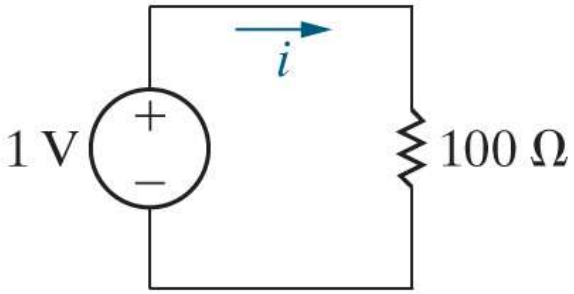
C. either series or parallel

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Chapter 3 – Simple resistive circuits
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If we want to measure the current *i* in the circuit shown here, we should insert an ammeter in _____ with the 100 Ω resistor.



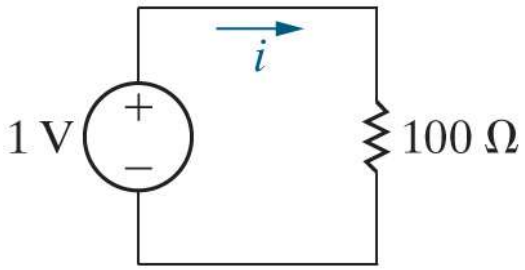
- ✔ A. series
- ✘ B. parallel
- ✘ C. either series or parallel

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Chapter 3 – Simple resistive circuits
(11 of 22)

If we insert an ideal ammeter in series with the $100\,\Omega$ resistor, what will the meter read?



A. 100 A

B. 1 mA

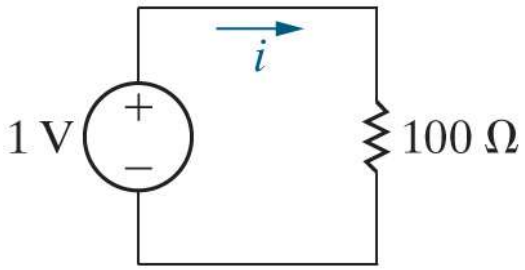
C. 10 mA

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Chapter 3 – Simple resistive circuits
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If we insert an ideal ammeter in series with the $100\,\Omega$ resistor, what will the meter read?



- ✗ A. 100 A
- ✗ B. 1 mA
- ✓ C. 10 mA

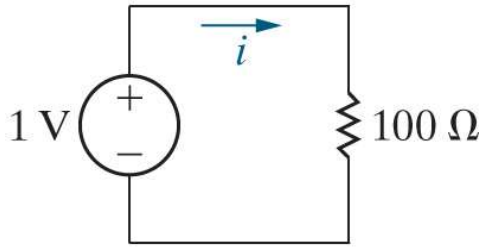
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Chapter 3 – Simple resistive circuits
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Measuring current and voltage

Suppose we use the ammeter from the example to measure the current *i* in the circuit here. What will the ammeter read?



The resistance of the ammeter is $5\,\Omega$. Since the ammeter is inserted in series with the $100\,\Omega$ resistor, the equivalent resistance seen by the 10 V source is $100 + 5 = 105\,\Omega$. Therefore,

$$i = \frac{1}{105} = 0.009524 = 9.524 \text{ mA}.$$

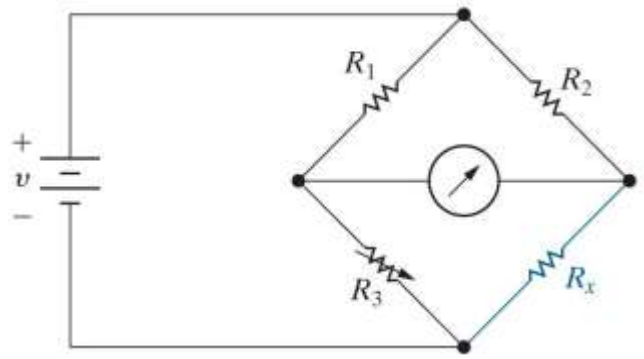
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Chapter 3 – Simple resistive circuits (13 of 22)

Measuring resistance

- An instrument that measures resistance is an ohmmeter. Many difference circuits can be used to measure resistance. Here we focus on the Wheatstone bridge. A Wheatstone bridge can accurately measure resistance in the range of $1\,\Omega$ to $1\text{M}\,\Omega$.
- A Wheatstone bridge consists of four resistors, a voltage source and a detector, which is usually a d'Arsonval movement in the μA range, called a galvanometer.



- The values of R_1 and R_2 are known.
- R_3 is a variable resistor, called a potentiometer, which can be set to a known value.
- The value of R_x is unknown, and is measured by the bridge circuit.

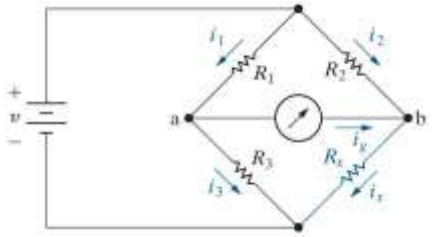
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Chapter 3 – Simple resistive circuits
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Measuring resistance

To find the unknown resistance R_x, adjust the R_3 resistor until there is no current in the galvanometer. Then the bridge is balanced and the unknown resistance is given by



$$R_x = \frac{R_2}{R_1} R_3$$

When the galvanometer current is 0

KCL at node a: $i_1 = i_3$; KCL at node b: $i_2 = i_x$

KVL in lower mesh (clockwise): $0 + R_x i_x - R_3 i_3 = 0 \quad \Rightarrow \quad R_x i_x = R_3 i_3$

KVL in upper mesh (clockwise): $R_2 i_2 - R_1 i_1 + 0 = 0 \quad \Rightarrow \quad R_2 i_2 = R_1 i_1$

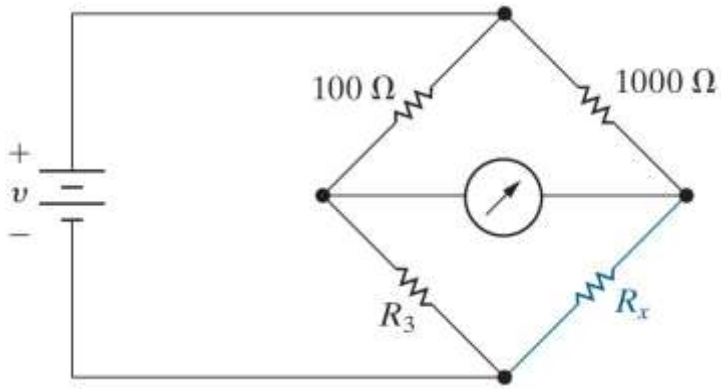
divide the two KVL equations: $\frac{R_x i_x}{R_2 i_2} = \frac{R_3 i_3}{R_1 i_1} \quad \Rightarrow \quad \frac{R_x}{R_2} = \frac{R_3}{R_1}$

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Chapter 3 – Simple resistive circuits
(16 of 22)

The bridge circuit here is balanced when $R_3 = 150\,\Omega$. What is the value of R_x?



A. $10\,\Omega$

B. $1500\,\Omega$

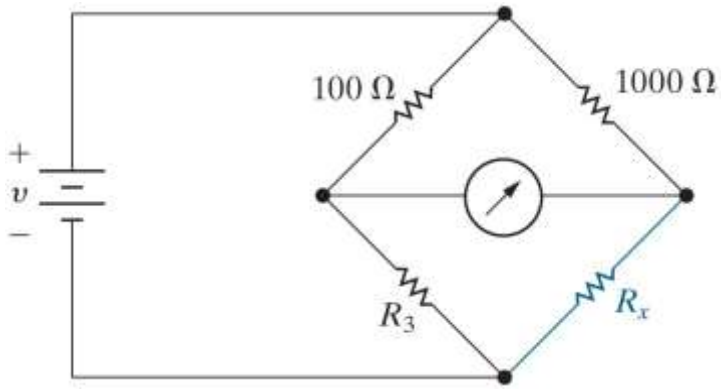
C. $100/15\,\Omega$

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Chapter 3 – Simple resistive circuits
(16 of 22)

The bridge circuit here is balanced when $R_3 = 150\,\Omega$. What is the value of R_x?



✗ A. $10\,\Omega$

✓ B. $1500\,\Omega$

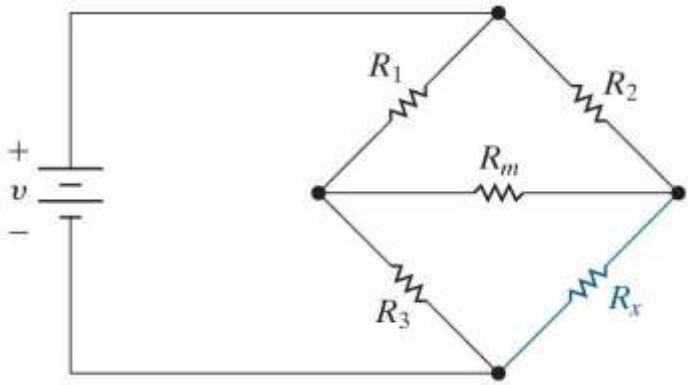
✗ C. $100/15\,\Omega$

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Chapter 3 – Simple resistive circuits
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Which statement is true about the resistors in this bridge circuit?



A. R_1 and R_3 are in series

B. R_m is in parallel with R_1 and R_2

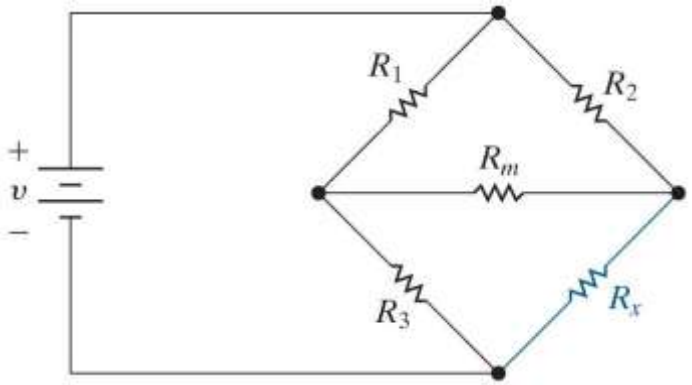
C. There are no parallel-connected or series-connected resistors

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Chapter 3 – Simple resistive circuits
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Which statement is true about the resistors in this bridge circuit?



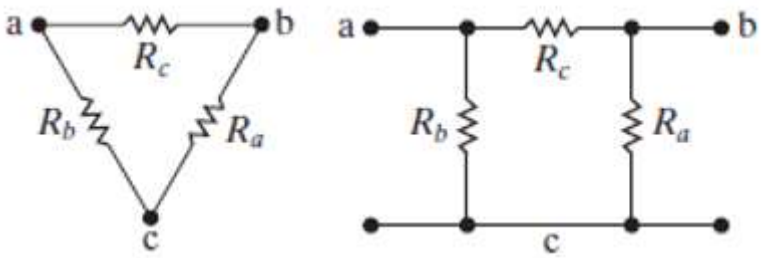
- ✗ A. R_1 and R_3 are in series
- ✗ B. R_m is in parallel with R_1 and R_2
- ✓ C. There are no parallel-connected or series-connected resistors

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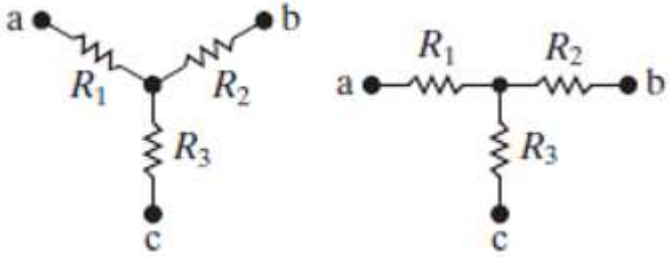
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Chapter 3 – Simple resistive circuits
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Delta-to-Wye (Pi-to-Tee) Equivalent Circuits
While the bridge circuit has no series-connected or parallel-connected resistors, it does have resistors connected in a delta (or pi) configuration and resistors connected in a wye (or tee) configuration.



A Δ configuration rearranged to a π

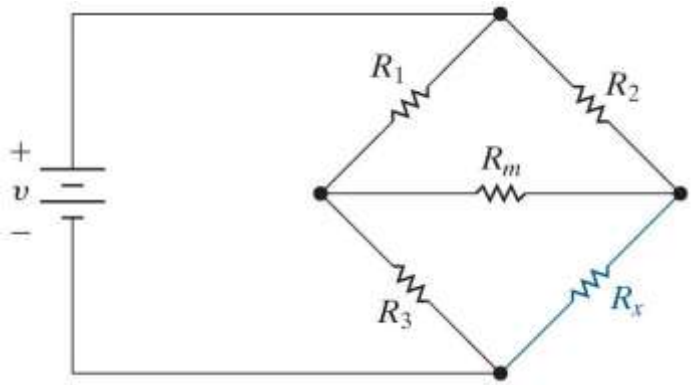


A Y configuration rearranged to a T

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Chapter 3 – Simple resistive circuits
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Which statement is false about the resistors in this bridge circuit?



A. R_1, R_2, R_m are connected in a delta

B. R_m, R_3 and R_x are connected in a wye

C. R_m, R_3 and R_x are connected in a delta

D. R_m, R_1 and R_3 are connected in a wye

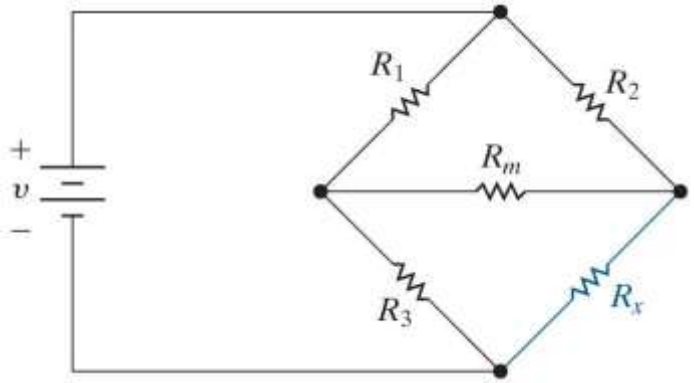
E. R_m, R_2 and R_x are connected in a wye

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Chapter 3 – Simple resistive circuits
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Which statement is false about the resistors in this bridge circuit?



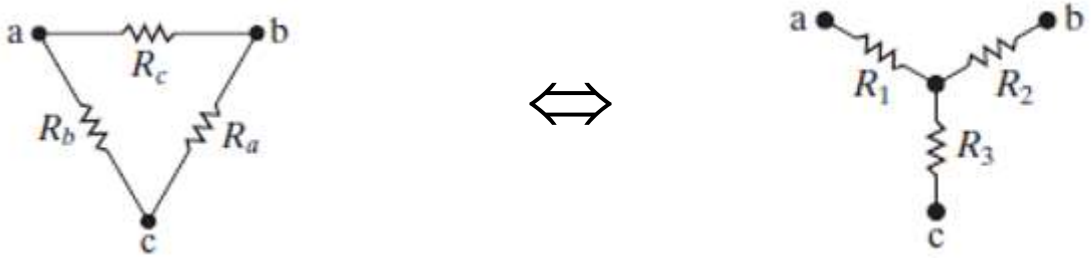
- ✗ A. R_1, R_2, R_m are connected in a delta
- ✓ B. R_m, R_3 and R_x are connected in a wye
- ✗ C. R_m, R_3 and R_x are connected in a delta
- ✗ D. R_m, R_1 and R_3 are connected in a wye
- ✗ E. R_m, R_2 and R_x are connected in a wye

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Chapter 3 – Simple resistive circuits
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Delta-to-Wye (Pi-to-Tee) Equivalent Circuits
A delta-connected configuration can be replaced by an equivalent wye-connected configuration, and vice versa.



$$R_a = \frac{R_1R_2 + R_2R_3 + R_1R_3}{R_1} \qquad R_1 = \frac{R_bR_c}{R_a + R_b + R_c}$$

$$R_b = \frac{R_1R_2 + R_2R_3 + R_1R_3}{R_2} \qquad R_2 = \frac{R_aR_c}{R_a + R_b + R_c}$$

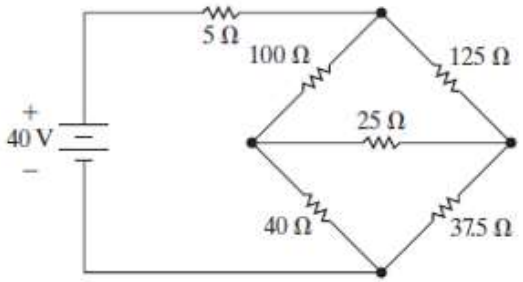
$$R_c = \frac{R_1R_2 + R_2R_3 + R_1R_3}{R_3} \qquad R_3 = \frac{R_aR_b}{R_a + R_b + R_c}$$

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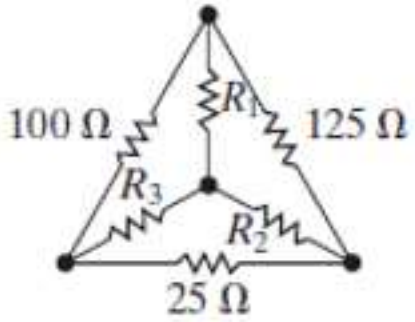
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Chapter 3 – Simple resistive circuits
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Delta-to-Wye (Pi-to-Tee) Equivalent Circuits – Example:
For the circuit shown here, find the current in the voltage source.



To do this, find the equivalent resistance seen by the voltage source. Start by replacing the upper delta with its equivalent wye.



$$R_1 = \frac{(100)(125)}{25+100+125} = 50\ \Omega$$

$$R_2 = \frac{(100)(25)}{25+100+125} = 12.5\ \Omega$$

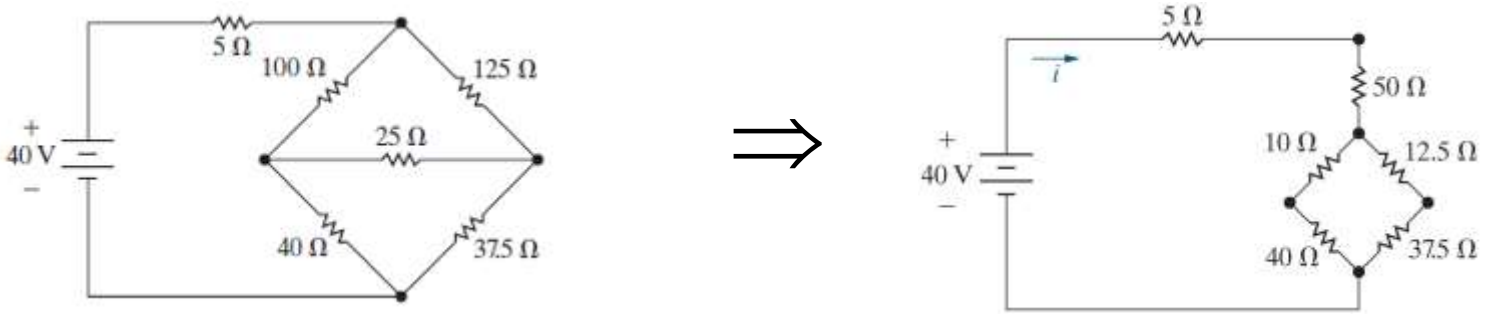
$$R_3 = \frac{(25)(100)}{25+100+125} = 10\ \Omega$$

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Delta-to-Wye (Pi-to-Tee) Equivalent Circuits – Example (continued):
For the circuit shown here, find the current in the voltage source.



The equivalent resistance seen by the source is

$$R_{eq} = 5 + 50 + \left[(10 + 40) \,||\, (12.5 + 37.5)\right] = 5 + 50 + \left[50 \,||\, 50\right]$$

$$= 5 + 50 + 25 = 80\ \Omega$$

Therefore, the current in the voltage source is

$$i = \frac{40}{80} = 0.5\ \text{A}.$$

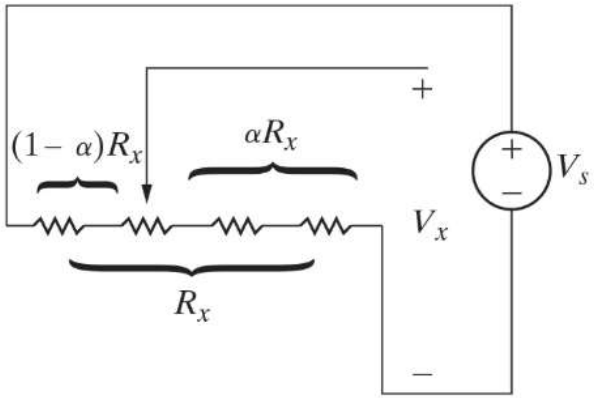
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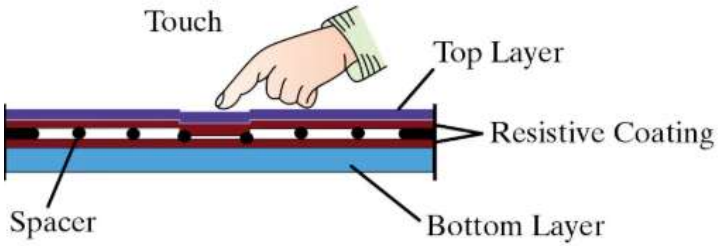
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$$V_x = \frac{\alpha R_x}{\alpha R_x + \left(1-\alpha\right)R_x} V_s = \frac{\alpha R_x}{R_x} V_s = \alpha V_s .$$

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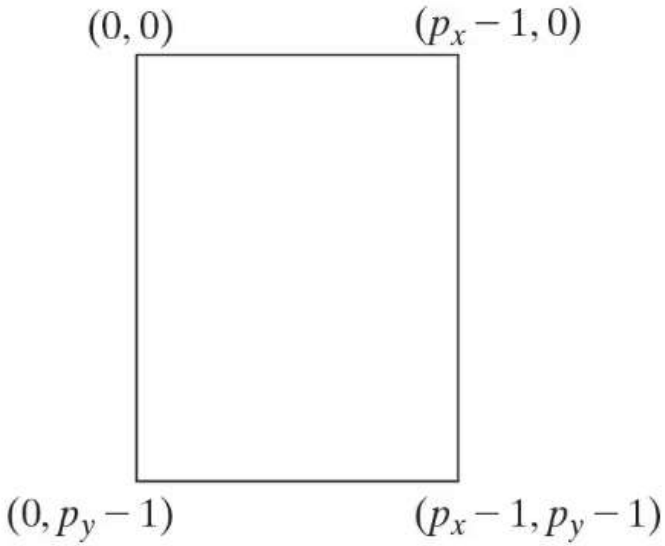
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$$V_x = \frac{\alpha R_x}{\alpha R_x + \left(1 - \alpha\right) R_x} V_s = \frac{\alpha R_x}{R_x} V_s = \alpha V_s.$$

$$\alpha = \frac{V_x}{V_s}.$$



$$x = (1 - \alpha)p_x.$$

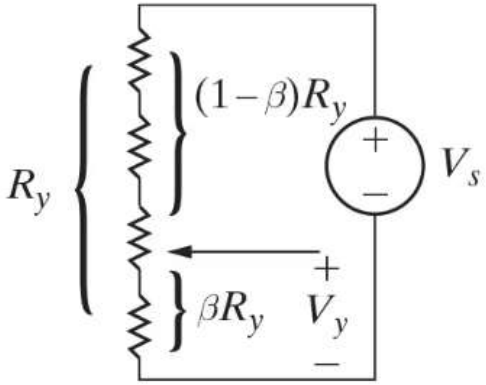
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$$V_y = \beta V_s.$$

$$y = (1 - \beta)p_y$$

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